

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

M1b: Summarizing Data

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College of Engineering

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Outline

- ① Survey Results
- ② Summary statistics
- ③ Quantiles and boxplots
- ④ Python/Colab
- ⑤ Outlook

Recap from Lecture 1a

- Two categories of uncertainty: aleatory and epistemic
- Visualize distributions via histograms and scattegrams
- Measures of location: mean, median, mode
- Measures of dispersion: range, variance, standard deviation, coefficient of variation

Objectives of today's lecture

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- Understand quantiles and boxplots

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- Learn how to use Python to generate descriptive statistics and graphical summaries of datasets (using Colab)

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$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (2)$$

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This is the square root of the variance:

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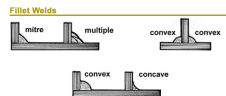
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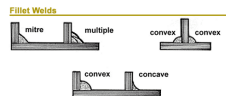
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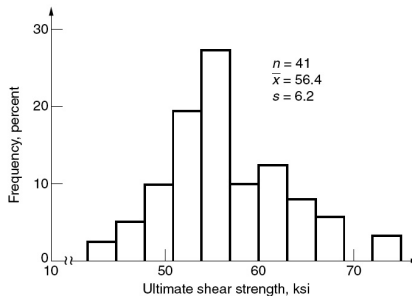
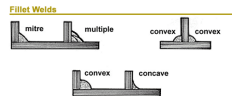
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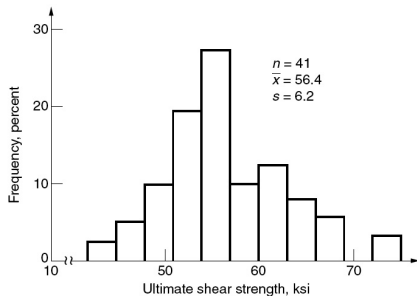
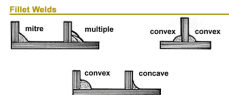
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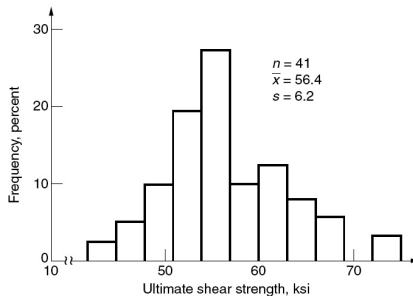
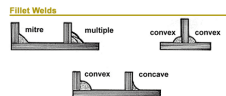
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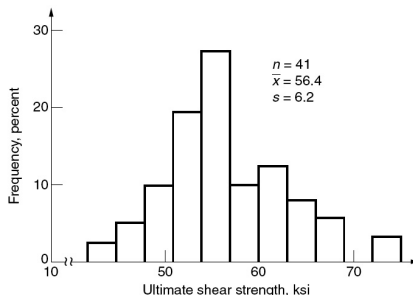
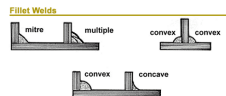
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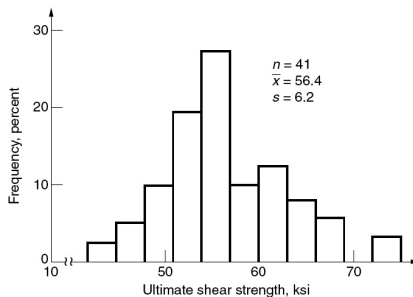
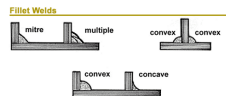
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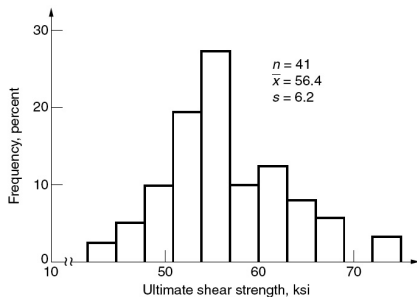
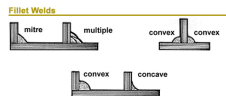
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Note: the mean and standard deviation have the same units

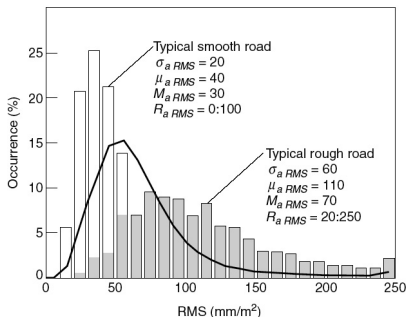
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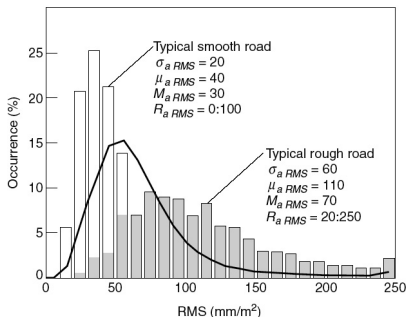
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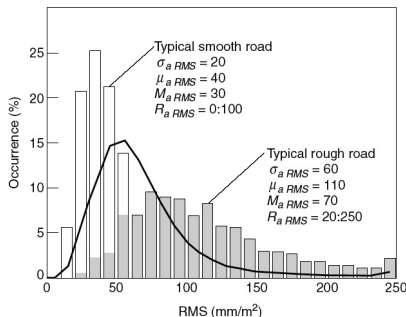


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- Which sample has the larger standard deviation?

Example 8: Comparing two distributions

Recall the surface roughness distributions from two samples:



Questions

- Which sample has the larger standard deviation?
- What are the implications?

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Measures how well the sample mean $\bar{x}$ estimates the population mean. Also known as **standard error**.

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Coefficient of variation

Measures dispersion relative to the mean.

$$\delta_X = \frac{s_X}{\bar{x}} \quad (6)$$

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- Ninety-nine **[per]centiles** split a sample into 100 groups: $P_1, \dots, P_{99}$

Quantiles (cont.)

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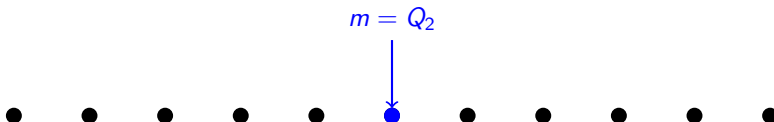
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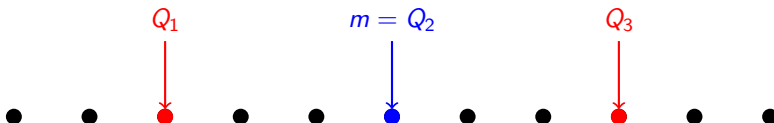
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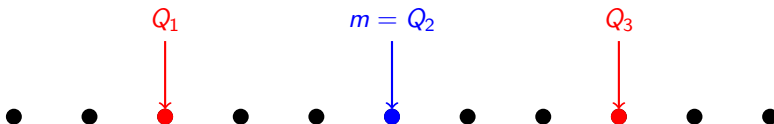
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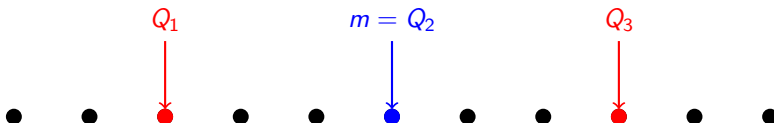
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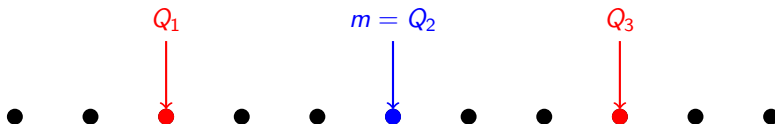
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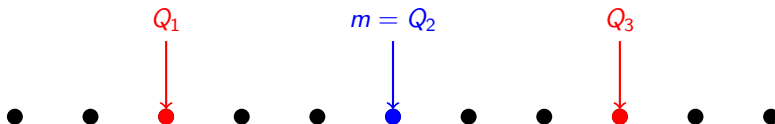


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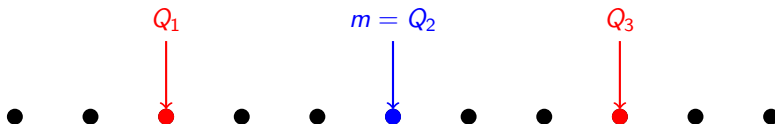


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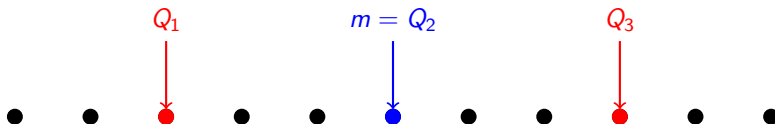
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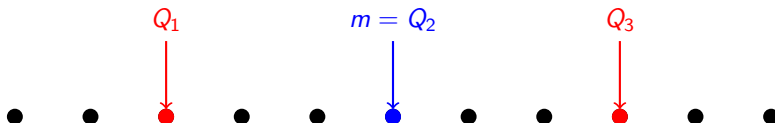
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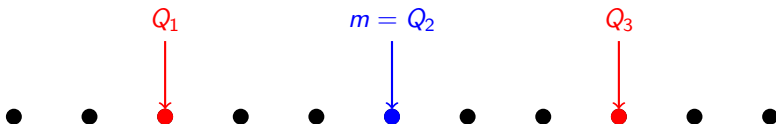
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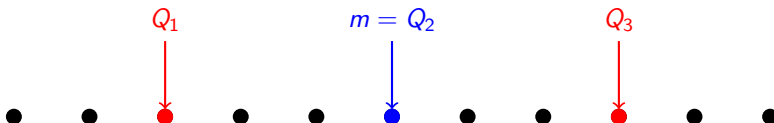
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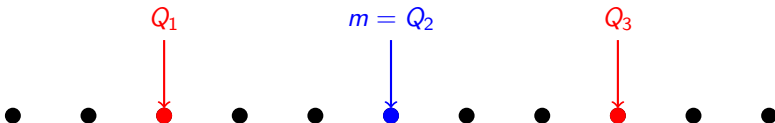
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- ② The second quartile Q_2 can be expressed as which percentile?^a

Answers: Q1: There are 4 quintiles; Q2: $Q_2 = P_{50}$ (fiftieth percentile)

^aRecall that a quartile splits a sample into 4 equal groups.

Using boxplots

¹Figure source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcd51>

Using boxplots

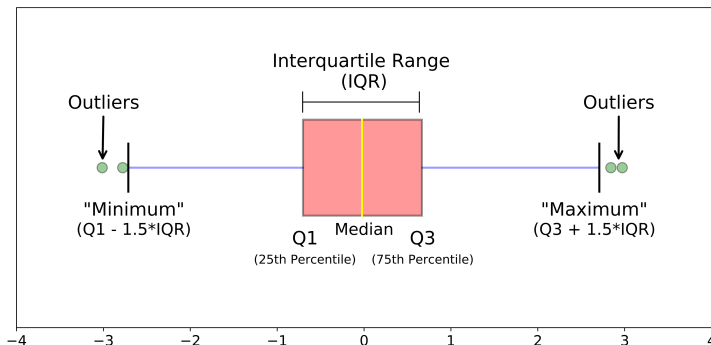
A boxplot¹ displays the distribution of data

¹Figure source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcd51>

Using boxplots

A boxplot¹ displays the distribution of data

- Useful for identifying outliers

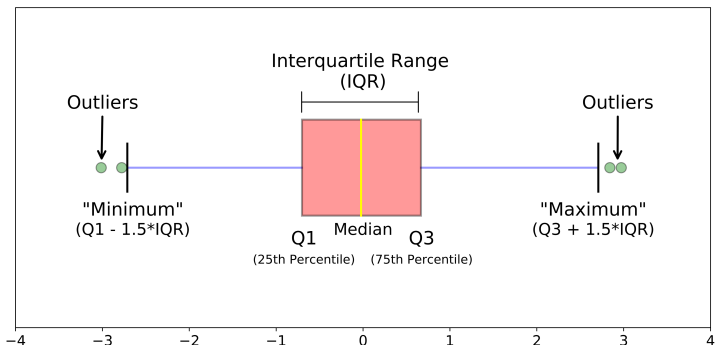


¹Figure source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcd51>

Using boxplots

A boxplot¹ displays the distribution of data

- Useful for identifying outliers
- Efficient for comparing multiple datasets

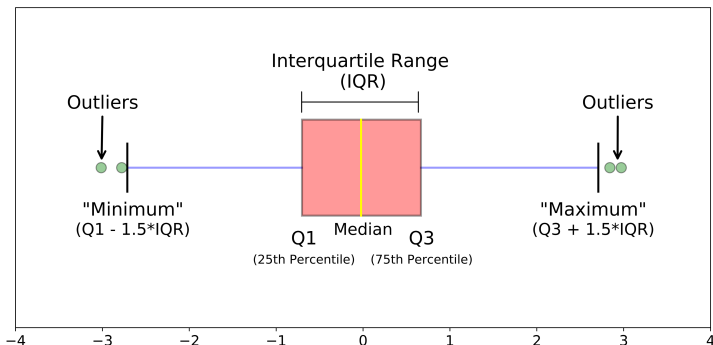


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Using boxplots

A boxplot¹ displays the distribution of data

- Useful for identifying outliers
- Efficient for comparing multiple datasets
- The lines indicating the “maximum/minimum” points (excluding outliers) are called *whiskers*



¹Figure source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcd51>

Python

Python

We will begin our introduction to Python via basic statistical analyses.
The platform will be Google Colab.

Summarizing data

Summarizing data

Example 10: Walking cadence

Summarizing data

Example 10: Walking cadence

In the article “Can We Really Walk Straight?” (*Amer. J. of Physical Anthropology*, 1992: 19–27) reported on an experiment in which each of 20 healthy men was asked to walk as straight as possible to a target 60m away at normal speed.

Summarizing data

Example 10: Walking cadence

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Consider the following observations on cadence (number of strides per second):

Summarizing data

Example 10: Walking cadence

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Consider the following observations on cadence (number of strides per second):

.95 .85 .92 .95 .93 .86 1.00 .92 .85 .81 .78 .93 .93 1.05 .93 1.06 .96 .81 .96

Summarizing data

Example 10: Walking cadence

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Consider the following observations on cadence (number of strides per second):

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Summarize the data; interpret and discuss.

Recap

- Pre-survey review

Recap

- Pre-survey review
- Python Introduction:

Recap

- Pre-survey review
- Python Introduction:
 - Summarizing data

Recap

- Pre-survey review
- Python Introduction:
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 - Visualizing data

Recap

- Pre-survey review
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 - Boxplots